Despeckling Images Using Elementary Cellular Automata

Swarna Aishwarya Twinkle(✉), Supreeti Kamilya, and Jit Mukherjee

Department of Computer Science and Engineering, Birla Institute Of Technology, Mesra, Ranchi, India
{phdcs10056.22,supreeti.kamilya17,jit.mukherjee}@bitmesra.ac.in

Abstract. Speckle noise significantly degrades the quality of images in domain especially in Synthetic Aperture Radar (SAR) and medical imaging, complicating the detection of critical features such as terrain details or tumors. The task of suppressing the speckle noise from images, known as *despeckling*, has significant importance in the field of image processing. In this study, we propose an innovative algorithm using Elementary Cellular Automata (ECAs) to reduce speckle noise. The methodology involves dividing an image into patches and converting pixel intensities within each patch into binary states. Using ECA rules, pixel values are iteratively adjusted to reduce noise while preserving details. To assess the effectiveness of our approach, we compare its performance with that of traditional filters using parameters like Peak-Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), Edge Preservation Index (EPI), and Perceptual Index (PI). The results of our ECA-based model achieved a PSNR of 28.532, surpassing traditional methods by significant margins: 8.1% higher than the Lee Filter, 12.8% above the Median Filter, 22.9% better than Anisotropic Diffusion, and over 57% higher than the Wiener Filter. The performance highlights the capability of our model in noise reduction compared to conventional approaches, offering a promising solution for applications in SAR and medical imaging.

Keywords: Elementary Cellular Automata · Speckle Noise · SAR Images · Ultrasound Images · PSNR

1 Introduction

Speckle noise remains a major challenge especially in satellite and medical images, often hiding critical details and compromising image clarity essential for accurate analysis. In satellite imaging, particularly with Synthetic Aperture Radar (SAR), speckle noise is a result of the interference from multiple scatterers within the observed area, making it challenging to detect fine details such as terrain features [13]. Similarly, in medical images such as in ultrasound, speckle appears due to the scattering of sound waves by tissue structures, which can hide key diagnostic features like tumors or blood vessels [14]. The removal of speckle noise from these images is crucial for improving the clarity of the data

S. Kamilya et al. (Eds.): ASCAT 2025, CCIS 2499, pp. 191–202, 2025.
https://doi.org/10.1007/978-3-031-94121-4_14

and enhancing the accuracy of analysis of scientific research as well as medical diagnosis. Various speckle suppression techniques such as Lee filter, Frost filter, Kuan filter [1], and Wiener filter [7] are commonly used to reduce speckle noise while preserving image edges and fine details. These filters adapt to the local variance of the image, smoothing the noise without significantly affecting important features. Deep learning technique, for example, DnCNN model [3], with its residual learning and batch normalization, is effective in handling Gaussian noise. Similarly, Feed Forward Network (FFDNET) [4] adapts to different noise levels by employing a tunable noise map, which improves denoising efficiency, especially down-sampled image segments. ID-CNN [5] uses convolutional layers, batch normalization, and residual connections to manage speckle noise effectively. Wavelet-based thresholding [6] works to separate noise from essential image structures, thereby retaining critical details that are crucial for diagnostic purposes. Anisotropic diffusion [8] reduces noise by selectively smoothing areas based on gradient information, which helps maintain edges and important tissue boundaries.

The discrete dynamical system, cellular automata (CAs) is used as effective tools in image processing due to their ability to model local interactions between neighboring cells. A recent research has explored the use of Cellular Automaton (CA) for image denoising, particularly through 2D-CA models [9]. These methods often work by treating image pixels as cells, where the state of each cell is influenced by its neighbors, allowing for noise suppression via local interactions [9]. While these approaches have shown good results in reducing noise, they primarily focus on other types of noise, such as Gaussian or Salt-and-Pepper noise. To the best of the authors' knowledge, there is no work on denoising speckle noise using 1-D cellular automata that leaves a gap in the existing literature for further exploration. Denoising such noise is challenging in image processing, especially in applications like remote sensing and medical imaging.

The present work uses the simplest kind of CAs, i.e., Elementary Cellular Automata (ECAs) for despeckling images. In this study, we propose an algorithm that uses ECA to reduce speckle noise, especially in SAR and medical images. We use SAR images and ultrasound breast cancer images from [15,16] and [17], that are prone to speckle noise. A controlled noise is introduced initially to the images. The images are then divided into small patches and average pixel intensity is calculated for each patch. The new pixel value is determined based on the pixel mean. Based on the new pixel value, a sum of neighborhood value is calculated. The next state value of the sum of an ECA rule decides if the pixel value of the image is to be incremented or decremented. This change in pixel values in the image reduces the speckle noise from the image. To evaluate its performance, we compare our proposed approach with traditional filters, assessing it on key parameters such as Peak-Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM), Edge Preservation Index (EPI), and Perceptual Index (PI). These comparisons allow us to measure the effectiveness of our denoising algorithm quantitatively while maintaining image quality, especially in the context of SAR and medical imaging.

2 Preliminaries

2.1 Elementary Cellular Automata

A Cellular Automaton (CA) is an autonomous finite state machine (FSM) which functions in discrete time and space. Each cell in a CA updates its state at every time step, based on a set of rules. The current state of a cell at time t (known as the present state, PS) is used to compute its state at the next timestep or the next state (NS).

In a CA with 3-neighborhood structure (cell i, its left neighbor, and its right neighbor), the next state S_i^{t+1} of the i^{th} cell is determined by:

$$S_i^{t+1} = f_i(S_{i-1}^t, S_i^t, S_{i+1}^t)$$

where S_{i-1}^t, S_i^t, and S_{i+1}^t are the present states of the left neighbor, the cell itself, and the right neighbor at time t. The configuration of states at time t, denoted by, $S^t = (S_1^t, S_2^t, \ldots, S_n^t)$, represents the present state of the entire CA with n cells. Each cell's next state function can also be represented as a truth table (Table 1), with the decimal equivalent of the outputs.

Elementary cellular automata are the simplest type of CA. They comprise of an array of cells, each of which can be in one of two possible states: 0 (white) or 1 (black) [11]. The state of each cell in the next generation/iteration is determined by its current state and the states of its immediate neighbors to the right and left. The rules governing this evolution can be represented by a lookup table (Table 1), specifying the next state of the cell based on the current configuration of itself and its two neighbors [11]. Since each neighboring cell configuration has three binary positions, there are $2^3 = 8$ possible combinations for these three cells. Each rule can be defined by assigning a 0 or 1 to each of these configurations, resulting in $2^8 = 256$ unique rules [10]. These rules are typically numbered from 0 to 255 in binary.

For example, "Rule 30" (Table 1) specifies that a cell will become 1 only in certain configurations, creating characteristic patterns over time. In our study we have worked with finite elementary cellular automata and used null boundary condition where left (right) neighbor of the leftmost (rightmost resp.) is null.

Table 1. Truth Table for Rule 30

PS	111	110	101	100	011	010	001	000	**Rule**
NS	0	1	1	1	1	0	1	0	30

2.2 Speckle Noise

Speckle noise is a prevalent form of interference that can reduce the image quality in both SAR and medical imaging. In SAR imagery, speckle occurs due to the coherent nature of radar waves, resulting in random intensity fluctuations throughout the image [13]. This type of noise emerges from the interference of scattered waves from numerous small scatterers within the scanned region, producing a textured pattern. Since speckle noise is multiplicative [20], it often hides finer details and complicates image analysis. In medical imaging, particularly ultrasound images, speckle acts as random pixel variations that blur the image, caused by sound wave scattering across different tissue structures. This can hinder essential features, such as tumors, blood vessels, or other pathologies [14], making it more difficult to diagnose and analyze conditions accurately. Figure 1 and 2 shows the speckled SAR and ultrasound image respectively.

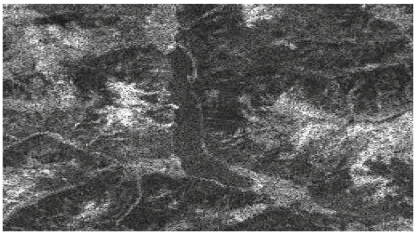

Fig. 1. Speckled SAR Image.

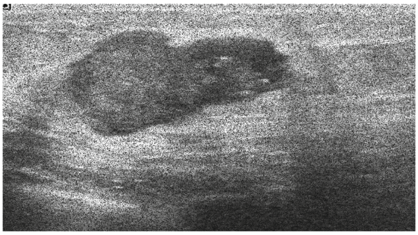

Fig. 2. Speckled Ultrasound Image.

3 Materials and Methods

In this study, we present a novel approach to denoise speckled images, specifically targeting SAR images and ultrasound breast cancer images. Speckle noise is a well-known challenge in satellite and medical imaging systems. By reducing this noise, we aim to enhance image quality making it easier to interpret and analyze crucial details in both remote sensing and medical applications. The following sections describe the data sources, preprocessing steps, and methodology used in this study.

3.1 Data

The ground truth dataset includes SAR images provided in papers [15] & [16] and ultrasound breast cancer images in [17], both of which are prone to speckle noise [18,19]. Speckle noise, which is multiplicative in nature [20], is a granular disturbance common in these types of images. It significantly impacts image clarity, making it difficult to interpret fine details. In the preprocessing step, we take clean images as ground truth images and add speckle noise to it. The introduction of controlled noise enables a direct assessment of denoising performance. The noisy image of size 256×256 is then divided into smaller non-overlapping patches of size 32×32 pixel. Dividing the image into smaller patches allows to address localized noise pattern within each region of the image, which results in effective denoising performance.

3.2 Methodology

As discussed, the image is first divided into smaller patches, and for each patch, the average pixel intensity is computed. This average intensity serves as a dynamic threshold for determining the new pixel value of each pixel within that patch. The new pixel value of each pixel is determined as follows:

$$\text{New Pixel Value} = \begin{cases} 1 & \text{if } I_{i,j} > \text{Patch Mean}, \\ -1 & \text{if } I_{i,j} < \text{Patch Mean}, \\ 0 & \text{if } I_{i,j} = \text{Patch Mean} \end{cases} \tag{1}$$

This conversion to new pixel values helps distinguish areas of higher intensity from areas of lower intensity, making it easier to apply the ECA rules effectively.

As an example, lets consider a 3×3 neighborhood with a central pixel (value 127) in a patch where the mean pixel intensity is calculated as 98. With the threshold of 98, the new pixel value for each pixel in the neighborhood are calculated following Eq. 1 as illustrated in Fig. 3.

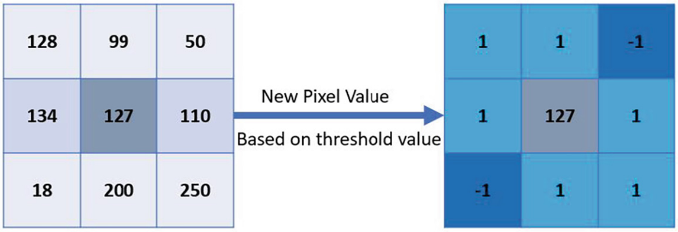

Fig. 3. Conversion to new pixel value based on threshold.

Here,

- Pixels with values greater than 98 will be assigned a new value of 1.
- Pixels with values less than 98 will be assigned a new value of –1.
- Pixels with values equal to 98 will be assigned a new value of 0.

The new pixel value of the neighboring pixels (excluding the central pixel) are summed to reflect the local contrast around the central pixel. The sum indicates whether the surrounding area has higher or lower intensities compared to the mean value. The sum of the neighborhood is computed as shown in Eq. 2

$$\text{Sum} = \sum_{k=-1}^{1} \sum_{l=-1}^{1} B_{i+k,j+l} - B_{i,j} \tag{2}$$

Here $B_{i,j}$ is the central pixel. Substituting the new pixel values of the example (shown in Fig. 3) in Eq. 2, we get,

$$Sum = (1) + (1) + (-1) + (1) + (1) + (-1) + (1) + (1) = 4 \tag{3}$$

This sum will then be used to determine the transformation of the central pixel. Once the sum is calculated, the next step is to apply the local transition of an ECA rule. The central pixel value is updated based on the sum, according to the following rules.

$$I_{i,j} = \begin{cases} I_{i,j} + 1 & \text{if Sum is positive\&Next State (NS) of the Sum is 1,} \\ I_{i,j} - 1 & \text{if Sum is negative\&Next State (NS) of the Sum is 1,} \\ I_{i,j} & \text{Next State (NS) of the Sum is 0} \end{cases} \tag{4}$$

These ECA-based rules are applied iteratively, allowing for the reduction of speckle noise while preserving the important features of the image. The more we get the difference while calculating the sum, the more change in color is there in a region. When the color is being denoised, it is checked whether any color is dominating the others. If the domination (i.e., the difference) is more, we smoothen it by incrementing or decrementing, depending on the sum value. Higher difference means a higher value of the ECA rule, based on whose NS the change is being made. Note that, the value of Sum can be 8 or –8 if all the neighbors of the central pixel are 1 or –1 respectively. This means, the central pixel is having different intensity where all values in the neighborhood are uniform. Therefore, we increment (decrement) the central pixel value if the Sum is 8 (–8 respectively).

As an example, let us consider ECA rule 245. It is represented in Table 2.

Table 2. Binary Input and Corresponding Output

111	110	101	100	011	010	001	000
1	1	1	1	0	1	0	1

In Eq. 3, we get the value of Sum as 4. We first convert 4 into binary representation of 3-bits i.e. 100. Now, according to Rule 245, next state of 100 is 1. Since the Sum in the equation is positive, so it indicates that the cental pixel should be incremented. Thus the central pixel 127 is changed to 128. This transformation process is applied to each patch across all patches, iteratively repeating over multiple iterations. Each pass applies ECA rules, allowing gradual noise reduction across the image. After processing, the denoised patches are reassembled into a single 256×256 image. Initially, all ECA rules i.e. 0 to 255 have been applied and tested. Out of which, few rules produced effective denoising results and are chosen for the method (provided in Sect. 4). The chosen rules show superior noise reduction while preserving image details, making them a good choice for this ECA-based denoising approach.

The steps of the methodology is summarized as follows.

1. Partition the image into non-overlapping $P \times P$ patches.
2. Calculate the average pixel intensity for each patch.
3. For each pixel in the patch, assign a new pixel value based on comparison with the patch mean, as given in Eq. 1.
4. Calculate Sum of the new pixel values of its 3×3 neighborhood, as shown in Eq. 2.
5. Based on the 3-bit binary representation of the sum, map it to the local transition of the ECA rule to determine if the central pixel should be incremented, decremented, or remain unchanged, as provided in Eq. 4.
6. Repeat Steps 3 to 5 for each patch, iterating over the entire image.
7. Combine all processed patches to reconstruct the final denoised image.

4 Results and Discussion

To thoroughly evaluate the performance of our method, we have tested it across all 256 possible rules. Out of these, specific rules such as 221, 223, 225, 227, 229, 231, 233, 235, 237, 239, 241, 245, 247, 249, and 253 gave notably high-quality denoising results. This is because, in our method, each patch's sum determines the transition applied to the central pixel by mapping it to a specific ECA rule. These rules assign a binary output, either 0 or 1, for each possible 3-bit pattern, which then governs whether the central pixel value is incremented, decremented, or remains unchanged.

Upon testing all 256 ECA rules, we observe that certain rules, particularly higher-value odd rules produced the most effective denoising results. This effectiveness arises because, in higher rules, the next state typically includes more 1 s, resulting in a higher incremental changes to pixel values across the image. This tendency towards the change contributes to improved noise reduction by actively adjusting pixel intensities in response to neighboring pixels. Conversely, in lower-value rules, the most common outcome of the next state is 0, leading to fewer pixel adjustments and a less pronounced denoising effect.

Rules with next state value as 0 for the 000 local transition (i.e., the even rules) generally prevent change in pixel values across the image. The unchanged pixel value indicates that the noise is not reduced. This explains the reason behind the effectiveness of odd rules in this study. Mostly odd rules greater than 221 are effective in this work. On the other hand, rule 255 (represented by 11111111) contains the maximum number of '1' as next state values. It proves less effective because it enforces changes even in areas that are already smooth, introducing excessive alterations that degrade image quality slightly. Thus, while Rule 255 can reduce noise, it also introduces unnecessary changes across the image, leading to over-processing. Therefore, our selected rules balance noise reduction with detail preservation, enabling our approach to achieve effective denoising without sacrificing important image features.

We compare our model with traditional filtering techniques, demonstrating its effectiveness in noise reduction. However, we did not extend the comparison to other deep learning (DL) models due to their higher time complexity. While typical DL models require 1 to 2 min per execution, the proposed ECA-based approach completes in just 5 to 10 s, achieving a speedup of approximately 85–95%. This emphasizes the potential of the proposed technique for real time processing. The significant reduction in execution time highlights the efficiency of our method. Further, training of complex networks, such as deep learning techniques, is time consuming and can have issues related to overfitting. The Fig. 4 illustrates the results achieved by applying various filters compared to our proposed method on SAR dataset. Figure 5 and 6 shows the performance of our model across multiple iterations (15, 25 and 40 iterations) on both datasets, highlighting the effectiveness of our method in reducing noise and enhancing image quality. At 15 iterations, noise is reduced but is still visible. By 25, features are clearer, and at 40, the image is smooth with well-defined edges. Beyond 40, PSNR stabilizes, indicating no further improvement in image quality.

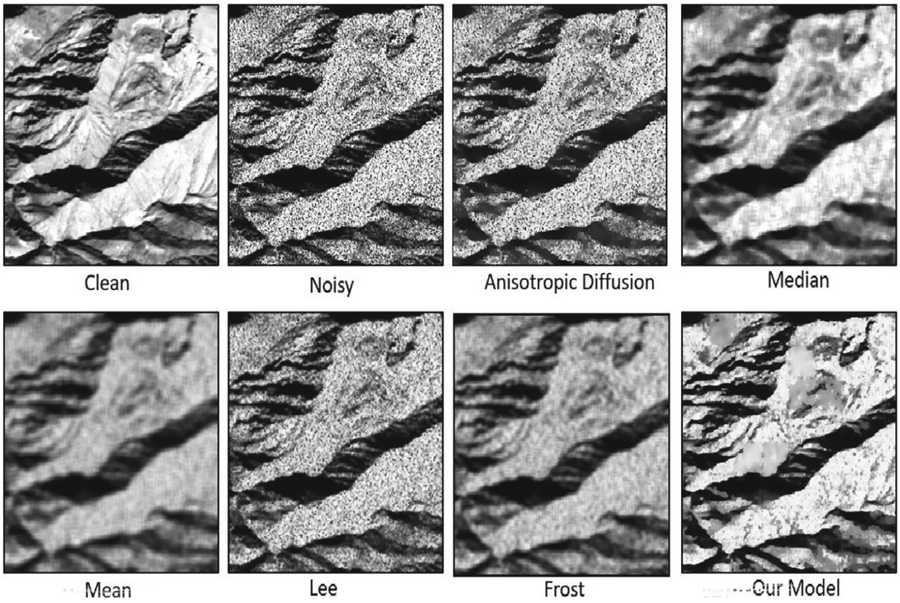

Fig. 4. Comparison with traditional filters.

To quantitatively assess the denoising effectiveness, we utilize the following image quality metrics.

1. *PSNR (Peak Signal-to-Noise Ratio)*: PSNR measures the ratio between the maximum possible signal power and the power of distorting noise. Higher PSNR values indicate better quality, indicating image denoising effectiveness.
2. *SSIM (Structural Similarity Index Measure)*: SSIM evaluates the similarity between the original and processed images by comparing luminance, contrast, and structure. It ranges from 0 to 1, where values closer to 1 indicate higher similarity and better image quality.

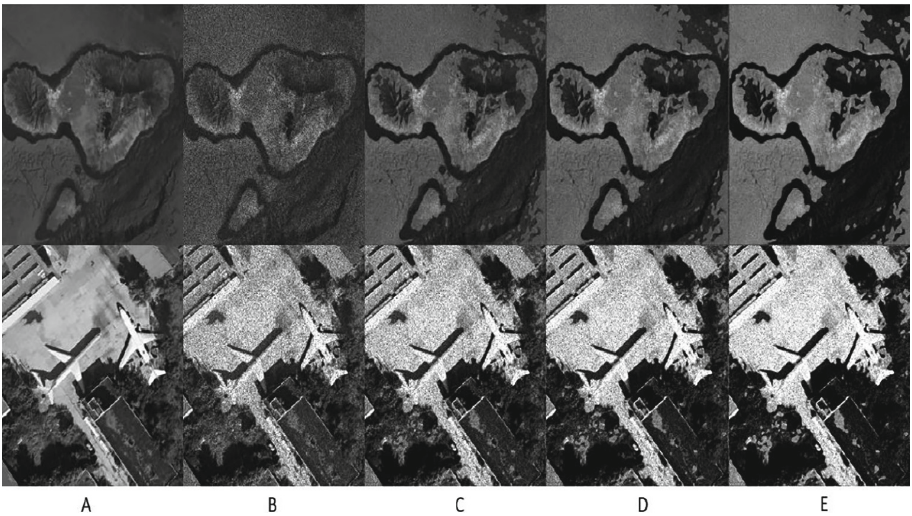

Fig. 5. Denoising Results on SAR Dataset; A: Clean Image; B: Noisy Image; C: After 15 iterations; D: After 25 iterations; E: After 40 iterations.

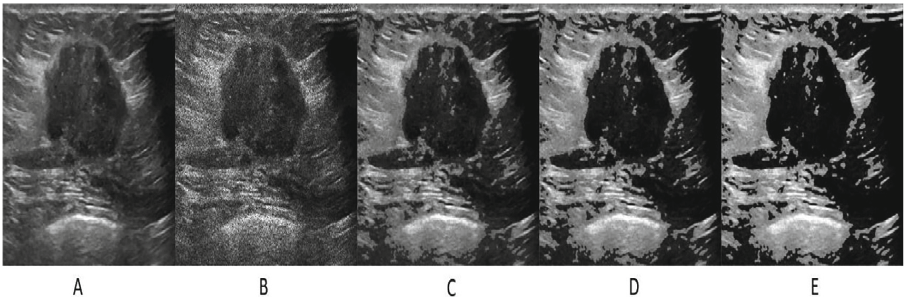

Fig. 6. Denoising Results on Ultrasound Dataset; A: Clean Image; B: Noisy Image; C: After 15 iterations; D: After 25 iterations; E: After 40 iterations

3. *EPI (Edge Preservation Index)*: EPI measures the effectiveness of the algorithm in preserving edges, an important factor in retaining image features during denoising. Higher EPI values means better edge preservation.
4. *PI (Perceptual Index)*: PI combines several perceptual quality measures to provide a more holistic view of image quality from a human perspective. Lower PI values indicate better perceived quality of the image.

By applying these metrics, we quantitatively validated that the selected rules performed well in enhancing image quality while effectively suppressing noise. Table 3 presents all the evaluation metrics used in this study. Graphical representations of these metrics are shown in Fig. 7.

Table 3. Performance Comparison of Speckle Suppression Filters

Method	PSNR	SSIM	EPI	PI
Our Model	**28.532**	**0.5872**	**0.5147**	**6.84**
Lee Filter	26.403	0.4733	0.3102	9.12
Median Filter	25.294	0.2058	0.1825	9.75
Anisotropic Diffusion	23.196	0.4303	0.2387	7.12
Mean Filter	22.804	0.4679	0.1457	7.93
Frost Filter	21.854	0.3493	0.2709	8.63
Wiener Filter	18.365	0.1368	0.2401	8.04
Noisy Image	12.117	0.3993	0.2473	7.52

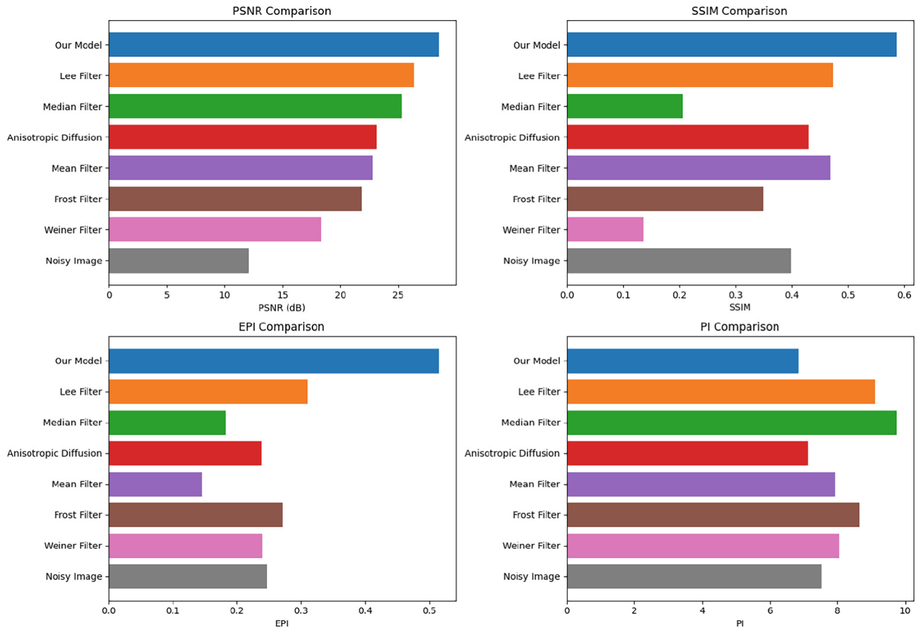

Fig. 7. Metrics (PSNR, SSIM, EPI and PI) Comparison of Our Model (Blue) with Lee Filter (Orange), Median Filter (Green), Anisotropic Diffusion (Red), Mean Filter (Purple), Frost filter (Brown), Wiener Filter (Pink), Noisy image (Grey). (Color figure online)

The potential limitation of this approach is using non-overlapping patches for denoising, as it can cause visible grid-like structure at the patch boundaries. To improve this, we aim to explore overlaps between patches to further reduce boundary artifacts and improve the overall denoising process. Additionally, we plan to implement non-uniform elementary cellular automata as future work, which has the potential to enhance both PSNR and image quality by adapting more effectively to different image structures.

5 Conclusion

This study presents a novel elementary cellular automata based approach for effective speckle noise reduction, especially important in SAR and medical imaging. By dividing the image into patches and applying ECA rules based on local transitions, our method achieves significant noise reduction while preserving essential image features. Through extensive testing, we identified specific ECA rules that enhance noise suppression, providing a balance between noise reduction and detail retention. Comparative results indicate that our model outperforms traditional filtering methods in terms of PSNR, SSIM, EPI and Perceptual Index. Our approach proves computationally efficient, requiring only 5 to 10 s per image. This quick processing time makes our approach practical for real-time applications, offering both quality and speed. Overall, our ECA-based approach provides a reliable, accessible alternative to more complex methods, making it a promising solution for applications that demand efficient, high-quality denoising.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

References

1. Singh, P., Diwakar, M., Shankar, A., Shree, R., Kumar, M.: A review on SAR image and its despeckling. Arch. Comput. Methods Eng. **28**(7), 4633–4653 (2021). https://doi.org/10.1007/s11831-021-09548-z
2. Baselice, F., Ferraioli, G., Johnsy, A.C., Pascazio, V., Schirinzi, G.: Speckle reduction based on wiener filter in ultrasound images. In: Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), pp. 3065–3068 (2015). https://doi.org/10.1109/EMBC.2015.7319039. PMID: 26736939
3. Zhang, K., Zuo, W., Chen, Y., Meng, D., Zhang, L.: Beyond a Gaussian denoiser: residual learning of deep cnn for image denoising. IEEE Trans. Image Process. **26**(7), 3142–3155 (2017). https://doi.org/10.1109/TIP.2017.2662206
4. Zhang, K., Zuo, W., Zhang, L.: FFDNet: toward a fast and flexible solution for CNN-based image denoising. IEEE Trans. Image Process. **27**(9), 4608–4622 (2018). https://doi.org/10.1109/TIP.2018.2839891
5. Wang, P., Zhang, H., Patel, V.M.: SAR image despeckling using a convolutional neural network. IEEE Signal Process. Lett. **24**(12), 1763–1767 (2017). https://doi.org/10.1109/LSP.2017.2756566
6. Sudha, S., Suresh, G.R., Sukanesh, R.: Speckle noise reduction in ultrasound images by wavelet thresholding based on weighted variance. Int. J. Comput. Theory Eng. (2009)
7. Shruthi, B., Renukalatha, S., Siddappa, M.: Speckle noise reduction in ultrasound images - a review. Int. J. Eng. Res. Technol. (IJERT) **4**(2) (2015)

8. Flores, W.G., de Albuquerque Pereira, W.C., Infantosi, A.F.C.: Breast ultrasound despeckling using anisotropic diffusion guided by texture descriptors. Ultrasound Med. Biol. **40**(11), 2609–2621 (2014). https://doi.org/10.1016/j.ultrasmedbio.2014.06.005
9. Chandra, N., Teotia, S., Chaudhary, A., Jain, V., Vashishtha, T., Dwivedi, N.N.: Noise reduction using cellular automata. Nanosci. Quant. Technol. **20**(6), 4650–4672 (2022). https://doi.org/10.14704/nq.2022.20.6.NQ22456
10. Kamilya, S., Adak, S., Das, S., Sikdar, B.: SACAs: (Non-uniform) cellular automata that converge to a single fixed point. J. Cell. Autom. **14**, 27–49 (2019)
11. Wolfram, S.: A New Kind of Science. Wolfram Media, 2002. ISBN 1579550088, https://www.wolframscience.com
12. Lee, J., Jurkevich, L., Dewaele, P., Wambacq, P., Oosterlinck, A.: Speckle filtering of synthetic aperture radar images: a review. Remote Sens. Rev. **8** (1994). https://doi.org/10.1080/02757259409532206
13. Kumar, L.A., Jebarani, M.E.: A comprehensive review on speckle denoising techniques in satellite images. In: 2019 International Conference on Communication, Control, and Signal Processing (ICCSP), pp. 245–248, 2019. https://doi.org/10.1109/ICCSP.2019.8697926
14. Pradeep, S., Nirmaladevi, P.: A review on speckle noise reduction techniques in ultrasound medical images based on spatial domain, transform domain and CNN methods. In: IOP Conference Series: Materials Science and Engineering, vol. 1055, no. 1, p. 012116, 2021. IOP Publishing
15. Schmitt, M., Hughes, L.H., Zhu, X.X.: The SEN1-2 dataset for deep learning in SAR-optical data fusion. In: ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences, vol. IV-1, pp. 141–146 (2018). https://doi.org/10.5194/isprs-annals-IV-1-141-2018
16. Dabhi, S., Soni, K., Patel, U., Sharma, P., Parmar, M.: Virtual SAR: A Synthetic Dataset for Deep Learning Based Speckle Noise Reduction Algorithms, 2020. arXiv preprint arXiv:2004.11021
17. Al-Dhabyani, W., Gomaa, M., Khaled, H., Fahmy, A.: Dataset of breast ultrasound images. Data Brief **28**, 104863. Elsevier (2020). https://doi.org/10.1016/j.dib.2019.104863
18. Singh, P., Shree, R.: Analysis and effects of speckle noise in SAR images. In: 2016 2nd International Conference on Advances in Computing, Communication, & Automation (ICACCA) (Fall), pp. 1–5 (2016). https://doi.org/10.1109/ICACCAF.2016.7748978
19. Li, X., Wang, Y., Zhao, Y., Wei, Y.: Fast speckle noise suppression algorithm in breast ultrasound image using three-dimensional deep learning. Front. Physiol. **13**, 880966 (2022). https://doi.org/10.3389/fphys.2022.880966
20. Lee, L., Jurkevich, P., Dewaele, P., et al.: Speckle filtering of synthetic aperture radar images: a review. Remote Sens. Rev. **8**(4), 313–340 (1994)